

	With same frequency as supply
6a	for a wave, electron can 'collect' energy continuously hence, electron will always be emitted / electron will be emitted at all frequencies after a sufficiently long delay
6bi	either wavelength is longer than threshold wavelength or frequency is below the threshold frequency or photon energy is less than work function
6bii	$hc / \lambda = \phi + E_{\text{MAX}}$ $(6.63 \times 10^{-34} \times 3.00 \times 10^8) / (240 \times 10^{-9}) = \phi + 4.44 \times 10^{-19}$ $\phi = 3.8 \times 10^{-19} \text{ J (allow } 3.85 \times 10^{-19} \text{ J)}$

6ci	photon energy larger so (maximum) kinetic energy is larger
6cii	fewer photons (per unit time) so (maximum) current is smaller
6di	light is re-emitted in all directions / only part of the re-emitted light is in the direction of the beam
6dii	an arrow between -3.40 eV and -1.51 eV and an arrow between -3.40 eV and -0.85 eV all arrows shown point 'upwards'
6diii	$E = hc / \lambda$ or $E = hf$ and $c = f\lambda$ $2.60 \times 1.60 \times 10^{-19} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$ $\lambda = 4.8 \times 10^{-7}$ m

7ai	Work is the product of a force and the distance moved in the direction of the force. $W = F \times \Delta r$
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